

# EXPLORAT

## Original P

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### Problem/Objective

#### The Exploration Barrier

Reinforcement learning (RL) methods perform well when **environments provide frequent reward signals**. However, many real-world and benchmark tasks contain sparse or delayed rewards, giving agents little feedback.

This makes **learning effective policies** through random exploration **extremely difficult**. As a result, agents often **fail to discover meaningful states or actions**, especially in complex environments like Montezuma's Revenge, where rewards require long sequences of correct actions. This exploration challenge limits the scalability of RL to large state spaces with sparse feedback.



# ION BY RANDOM

## Poster Presentation

### Paper Authors

Edwards, Amos Storkey, Oleg Klimov

### Methodology

#### Intrinsic Reward

In sparse-reward environments, RL agents may receive little to no feedback from environment.

To encourage exploration, an intrinsic reward component is added to the reward signal.

Total reward becomes the sum of extrinsic ( $e_t$ ) and intrinsic reward ( $i_t$ ):

$$r_t = e_t + i_t$$

$i_t$  is large for unfamiliar states and decreases as states become familiar.

#### Random Network Distillation (RND)

Intrinsic reward is computed as the MSE

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## Results

### Montezuma's Revenge:

D established a new state-of-the-art. It consistently found over half the rooms (22 out of 40) and occasionally completed the first level, achieving scores over 17,500.

### Comparison to Baselines:

D significantly outperformed standard PPO and "dynamics-based" (forward-prediction) exploration games like Venture and Private Eye.

### Curiosity Exploration:

Even when the game score was completely ignored, the RND agent successfully explored 21 rooms in Montezuma's Revenge just out of "curiosity".

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## Limitations

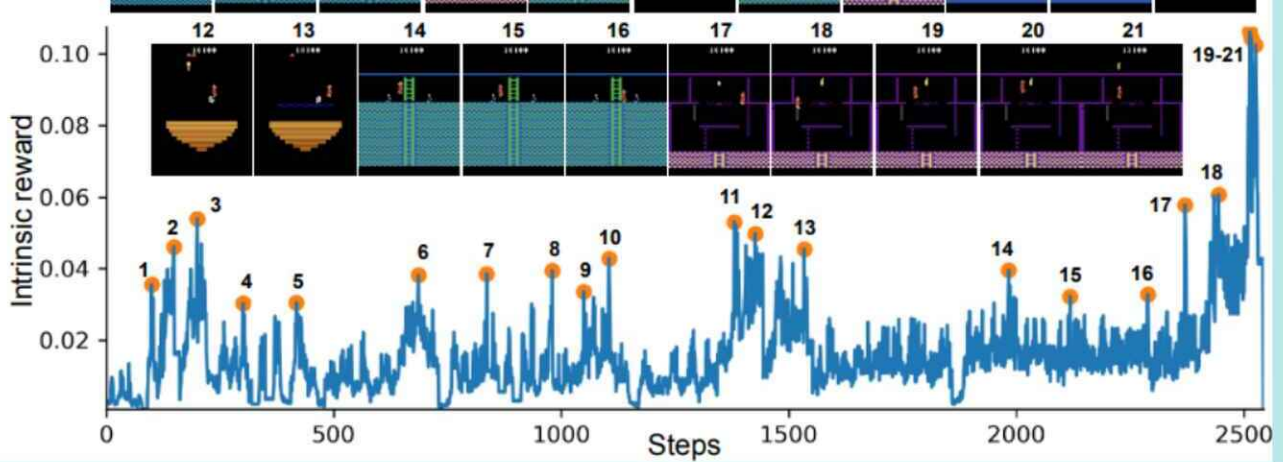
**Representation Sensitivity:** RND's novelty bonus depends heavily on the random feature space. Poor CNN features can make unrelated states appear similar, causing the agent to miss important distinctions.

**Reward non-stationarity :** As the predictor improves, the intrinsic reward signal shrinks over time, which can destabilize training if not properly normalized.

**No environment model :** RND has no understanding of why a state is novel, only that it is. It can get stuck exploring usually complex but strategically irrelevant states.

## Why It Matters?

directly addresses one of the hardest open problems in RL: learning useful behavior when rewards are



## Rewarding Novel States

To address this challenge, the paper proposes **Random Network Distillation (RND)**, an intrinsic motivation method that encourages agents to **explore unfamiliar states**.

The approach introduces an exploration bonus based on prediction error between **two neural networks**: a fixed randomly initialized target network and a predictor network trained to imitate it.

When the agent encounters a new or rarely visited state, the predictor network produces a **large prediction error**, generating a **high intrinsic reward**. As the agent revisits the state, the prediction error decreases and the reward fades. This mechanism allows the agent to naturally **prioritize novel states**, improving exploration efficiency and enabling **learning in environments with sparse external rewards**.

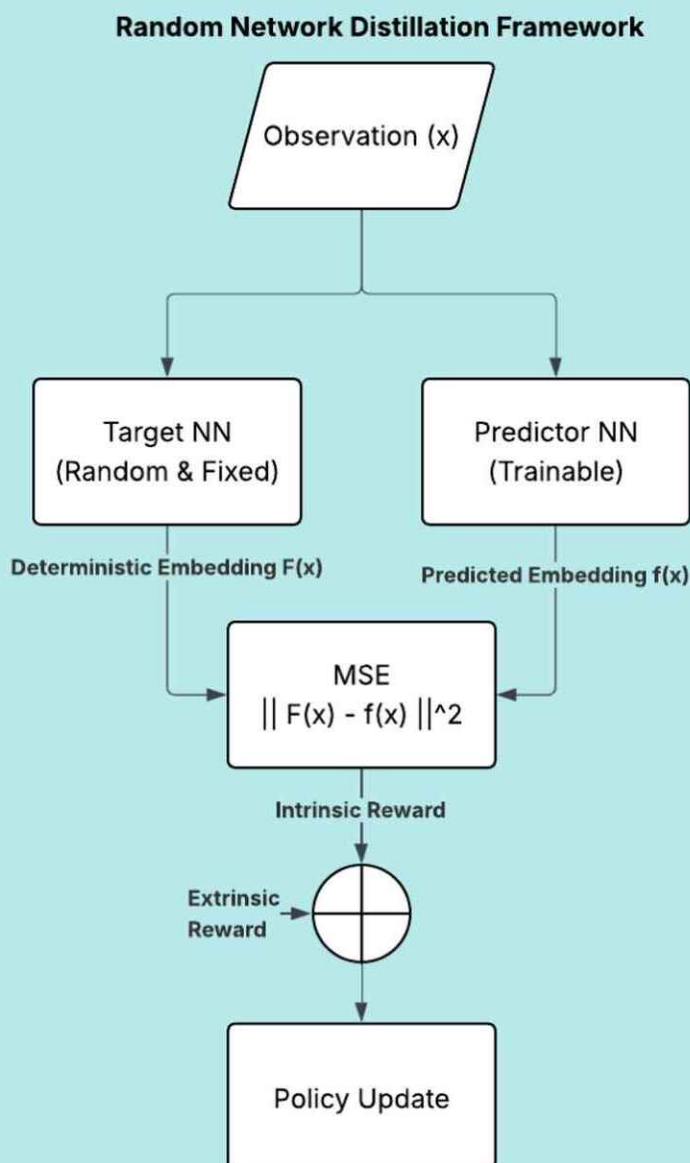
### **Objective:**

Develop a scalable exploration strategy that encourages agents to visit novel states using intrinsic rewards derived from prediction error.

between two neural networks:

- **Target Network:** Randomly initialized & fixed.
- **Predictor Network:** Tries to imitate the target.

When encountering a new state, the predictor network produces a large error  $\rightarrow$  Strong intrinsic reward. As a state is revisited, the predictor network learns them, the MSE decreases, and the reward decreases.



### **non-episodic intrinsic rewards improve exploration:**

When combining intrinsic + extrinsic, using non-episodic intrinsic (curiosity doesn't "die" at game over) increases number of rooms explored (especially for RNN policies).

### **Scale helps (many parallel environments):**

With more parallel environments (larger experience batches), performance improves—policy learns to exploit transient intrinsic rewards before they vanish. Paper shows very high returns and deep exploration at scale; in some runs the agent visits 24 rooms of level 1 and sometimes passes the next level.

### **Qualitative behavior:**

Agent sometimes seeks dangerous, rare states (jumping around hazards), such states remain underrepresented in experience, so they keep producing intrinsic reward.

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## **Key Sources & Acknowledgements**

Burda, Y., Edwards, H., Storkey, A., & Klimov, O. (2019). Exploration without counts. [arXiv](#)

Schulman, J., et al. (2017). Proximal Policy Optimization Algorithms. [arXiv](#)

Johnny Code. Reinforcement Learning Tutorials. YouTube. [youtube.com/johnnycode](#)

RL. learning useful behavior when **rewards** are **extremely rare**. It achieves superhuman-level performance in Montezuma's Revenge – a benchmark that had stumped RL researchers for years – using a surprisingly simple mechanism. Beyond games, this has real implications for robotics, scientific discovery, and any domain where trial-and-error exploration is costly and feedback is sparse.

## Conclusion

It offers an elegant, scalable solution to the hard exploration problem by repurposing **prediction error** as a **curiosity signal**. By keeping the target network fixed and the source network updated, it avoids the noisy-TV pitfall that undermines model-based curiosity methods. The dual reward stream – combining episodic extrinsic returns with non-episodic intrinsic returns – proves critical for crossing episode boundaries and achieving deep exploration. Its simplicity and compatibility with standard RL algorithms like PPO make it a strong baseline for any sparse-reward setting.

Exploration by Random Network Distillation. ICLR 2019.

Algorithms. arXiv:1707.06347

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